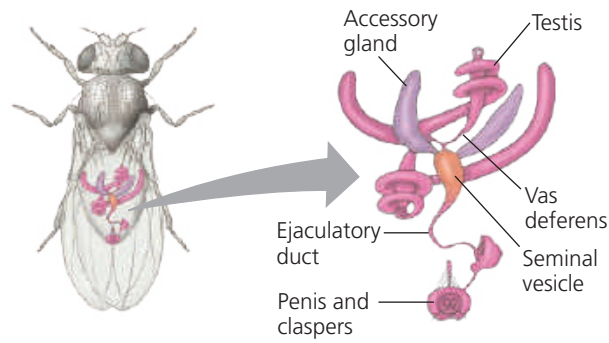
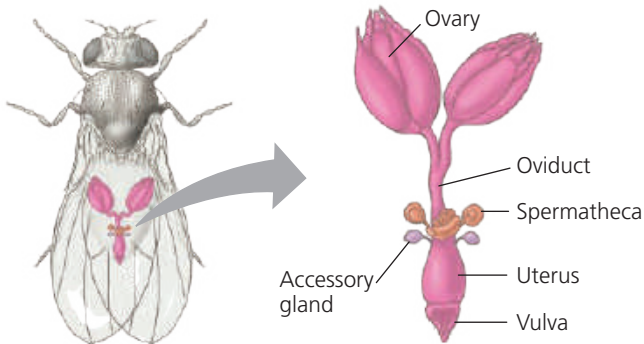


▼ **Figure 46.7** An example of insect reproductive anatomy.



(a) **Male fruit fly.** Sperm form in the testes, pass through a sperm duct (vas deferens), and are stored in the seminal vesicles. The male ejaculates sperm along with fluid from the accessory glands. (Males of some species of insects and other arthropods have appendages called claspers that grasp the female during copulation.)



(b) **Female fruit fly.** Eggs develop in the ovaries and then travel through the oviducts to the uterus. After mating, sperm are stored in the spermathecae, which are connected to the uterus by short ducts. The female uses a stored sperm to fertilize each egg as it enters the uterus before she passes the egg out through the vulva.

VISUAL SKILLS Study the two drawings, and then describe the movement of fruit fly sperm from formation to fertilization.

complex reproductive systems (Figure 46.7). In many insect species, the female reproductive system includes one or more **spermathecae** (singular, *spermatheca*), sacs in which sperm may be kept alive and stored for extended periods, a year or more in some cases. Because the female releases male gametes from the spermathecae and thus fertilizes her eggs only in response to the appropriate stimuli, fertilization occurs under conditions likely to be well suited to survival of offspring.

Vertebrate reproductive systems display limited but significant variations. In some vertebrates, the uterus is divided into two chambers; in others, including humans and birds, it is a single structure. In many nonmammalian vertebrates, the digestive, excretory, and reproductive systems have a common opening to the outside, the **cloaca**, a structure probably present in the ancestors of all vertebrates. Lacking a well-developed penis, males of these species instead release sperm by turning the cloaca inside out. In contrast, mammals generally lack a cloaca and have a separate opening for the digestive tract. In addition, most female mammals have separate openings for the excretory and reproductive systems.

Although fertilization involves the union of a single egg and sperm, animals often mate with more than one member

of the other sex. Monogamy, the sustained sexual partnership of two individuals, is rare among animals, including most mammals. Mechanisms have evolved, however, that enhance the reproductive success of a male with a particular female and diminish the chance of that female mating successfully with another partner. For example, some male insects transfer secretions that make a female less receptive to courtship, reducing the likelihood of her mating again.

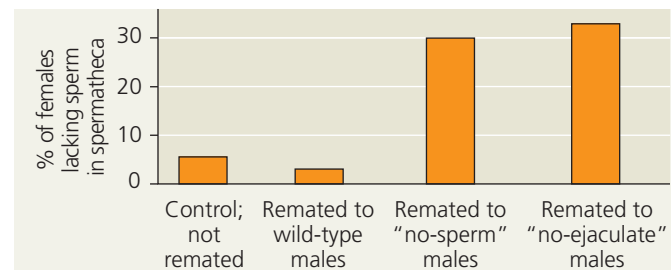
Can females also influence the relative reproductive success of their mates? This question intrigued two scientific collaborators working in Europe. Studying female fruit flies that copulated with one male and then another, the researchers traced the fate of sperm transferred in the first mating. As shown in Figure 46.8, females play a major role in determining the outcome of multiple matings. The processes by which gametes and individuals compete during reproduction remain a vibrant research area.

▼ **Figure 46.8** Inquiry

Why is sperm usage biased when female fruit flies mate twice?

Experiment When a female fruit fly mates twice, 80% of the offspring result from the second mating. Scientists had hypothesized that ejaculate from the second mating displaces sperm from the first mating. To test this hypothesis, Rhonda Snook, at the University of Sheffield, and David Hosken, at the University of Zurich, used mutant males with altered reproductive systems. “No-ejaculate” males mate but do not transfer sperm or fluid to females. “No-sperm” males mate and ejaculate but make no sperm. The researchers allowed females to mate first with wild-type males and then with wild-type males, no-sperm males, or no-ejaculate males. As a control, some females were mated only once (to wild-type males). The scientists then dissected each female under a microscope and recorded whether sperm were absent from the spermathecae, the major sperm storage organs.

Results



Conclusion Because remating reduces sperm storage when no sperm or fluids are transferred, the hypothesis that ejaculate from a second mating displaces stored sperm is incorrect. Instead, it appears females may get rid of stored sperm in response to remating, perhaps allowing for replacing stored sperm, possibly of diminished fitness, with fresh sperm.

Data from R. R. Snook and D. J. Hosken, Sperm death and dumping in *Drosophila*, *Nature* 428:939–941 (2004).

WHAT IF? Suppose males in the first mating had a mutant allele that resulted in smaller eyes as a dominant trait. What fraction of the females would produce some offspring with smaller eyes?